1.10 Quiz: Arithmetic and Geometric Sequences — Markscheme

Calculator allowed. Total: 34 marks.

1. [Maximum mark: 8]

   15, 23, 31, 39, ...

   (a) $d = 23 - 15 = 8$ **A1**

   (b) $u_n = u_1 + (n-1)d$ **(M1)**

   $u_n = 15 + 8(n-1)$ **A1**

   Accept $u_n = 8n + 7$, or any equivalent expression, for full marks. A candidate who writes $u_n = 15 + 8n$ earns M1A0. The general formula $u_n = u_1 + (n-1)d$ with no values substituted earns M0A0. A substituted expression that appears only in a later part earns both marks here, even one for a single value of n ("$15 + (30-1)8$" in (c) shows $u_1 = 15$ and $d = 8$ substituted).

   FT: award A1 for a correct expression from the candidate's own d in (a).

   (c) $u_{30} = 15 + 8(30-1)$ **(M1)**

   $= 15 + 232 = 247$ **A1**

   A common error is $15 + 8(30) = 255$: M0A0 (it is u_{31}). FT from the candidate's expression in (b).

   (d) $15 + 8(n-1) = 303$ **M1**

   $8(n-1) = 288$

   $n - 1 = 36$

   $n = 37$ **A1**

   n is a positive whole number, so 303 is a term: the 37th term. **R1**

   Note: the statement that $n = 37$ is a whole number (or an equivalent, such as "n is a positive integer") is required for R1. A conclusion alone, such as "$303 = u_{37}$", earns R0.

   Accept $8n + 7 = 303$, $8n = 296$, $n = 37$. R1 requires the conclusion linked to n being a whole number; a bare "$n = 37$" earns M1A1R0. Accept a table search that shows $u_{37} = 303$ for full marks.

2. [Maximum mark: 9]

$u_5 = 38,\ \ u_{14} = 11.$

(a) $u_1 + 4d = 38 \quad$ and $\quad u_1 + 13d = 11$ **M1**

$9d = 11 - 38 = -27$

$d = -3$ **A1**

$u_1 = 38 - 4(-3) = 50$ **A1**

M1 for any valid route: both equations, $d = \dfrac{u_{14} - u_5}{14 - 5}$, or solving the system on the GDC. The two A1 marks are independent. FT: award the second A1 for u_1 correct from the candidate's own d.

(b) $50 - 3(n-1) < 0$ **M1**

$3(n-1) > 50$

$n - 1 > 16.67$

$n > 17.67$, so $n = 18$ **A1**

$u_{18} = 50 - 3(18-1) = -1$ **A1**

The question asks for the term, not n: the final A1 is for -1. A final answer of "$n = 18$" alone, or a named term with no value ("u_{18}"), earns M1A1A0. A bare "$u_{18} = -1$", or even "-1", earns full marks: the step is short enough that no working is needed.

Accept an equation in place of the inequality, or a table search showing $u_{17} = 2$ and $u_{18} = -1$ (markscheme-spec §18). FT from the candidate's u_1 and d.

(c) Equal differences: $(4k-3) - (2k+1) = (5k+1) - (4k-3)$ **M1**

$2k - 4 = k + 4$

$k = 8$ **A1**

First three terms: 17, 29, 41 $\quad (d = 12)$ **A1**

Accept the equivalent set-up $2(4k-3) = (2k+1) + (5k+1)$ (the middle term is the mean of its neighbours). FT: award the final A1 for three terms correctly evaluated from the candidate's k.

3. [Maximum mark: 10]

(a)

(i)	5, 10, 20, 40	geometric, $r = 2$	**A1**
(ii)	7, 4, 1, -2	arithmetic, $d = -3$	**A1**
(iii)	2, 3, 5, 8	neither	**A1**
(iv)	162, 54, 18, 6	geometric, $r = \frac{1}{3}$	**A1**

Each A1 needs the type written as a word ("arithmetic", "geometric", "neither") and, for (i), (ii) and (iv), the correct d or r. A value alone ("$r = 2$", "$\times 2$") does not name the type: A0. With the word present, accept r written as an operation ("$r = \times 2$", "$r = \div 3$"), with a feedback note to write r as a number. Accept $r = 0.333$ in (iv). In (iii) the differences are $1, 2, 3$ and the ratios $1.5, 1.67, 1.6$, so neither is constant. $d = 3$ in (ii) (sign lost) earns A0.

(b) $r = \dfrac{20}{16} = 1.25$ **A1**

(c) $u_{12} = 16(1.25)^{12-1}$ **(M1)**

$= 16(1.25)^{11} = 186 \quad (186.264\ldots)$ **A1**

A1 for 186 *(3 s.f.) or the calculator display written with an ellipsis (*$186.264\ldots$, $186.26\ldots$*). A value cut short with no ellipsis (*186.26, 186.265, 186.3*, or the full display* 186.2645149 *written without one) earns A0 (markscheme-spec §21).* $16(1.25)^{12} = 233$ *is* u_{13}*: M0A0. FT from the candidate's r in (b).*

(d) From u_3 to u_6 is 3 steps of $\times r$: $r^3 = \dfrac{1215}{45} = 27$ **M1**

$r = 3$ **A1**

$u_1 = \dfrac{u_3}{r^2} = \dfrac{45}{9} = 5$ **A1**

Note: "find" needs supporting working. Bare answers $r = 3$, $u_1 = 5$ with no working earn 1 mark (M0A1A0), never full marks (Chris, 2026-10-18).

Accept the equations $u_1 r^2 = 45$ and $u_1 r^5 = 1215$, divided to give $r^3 = 27$, for the M1. A candidate who uses $u_1 = \dfrac{u_3}{r} = 15$ earns M1A1A0. FT: award the final A1 for u_1 correct from the candidate's own r.

4. [Maximum mark: 7]

Job A: arithmetic, $u_1 = 38\,000$, $d = 2000$. Job B: geometric, $u_1 = 36\,000$, $r = 1.06$.

(a) Job A: $A_n = 38\,000 + 2000(n-1)$ **A1**

Job B: $B_n = 36\,000(1.06)^{n-1}$ **A1**

Accept $A_n = 36\,000 + 2000n$. *A common error is* $r = 0.06$ *or* $r = 6$ *in Job B: A0.* $B_n = 36\,000(1.06)^n$ *earns A0 (it gives* \$38 160 *in year* 1*).*

(b) Job A: $A_3 = 38\,000 + 2000(3-1) = \$42\,000$ **A1**

Job B: $B_3 = 36\,000(1.06)^{3-1} = \$40\,449.60$ **A1**

Accept \$40 400 *(3 s.f.). FT: award each A1 for the value from the candidate's own expression in (a). The precision rule applies to FT values too: an own value cut short (*\$42 876 *from* 42 876.576*) earns A0.*

(c) Compare the two salaries year by year: **M1**

Year 6 **A1**

Supporting work: **A1**

Year	Job A	Job B
5	\$46 000	\$45 449.17
6	\$48 000	\$48 176.12

M1 for a valid comparison of the two expressions: a GDC table, a graph, or solving $36\,000(1.06)^{n-1} = 38\,000 + 2000(n-1)$ *on the GDC. First A1 for year* 6*. Second A1 for supporting work that shows year* 6 *is the first: a table with both the year-*5 *and year-*6 *rows, or a sketch of the two graphs intersecting, with the intersection marked by its value (*$n = 5.78\ldots$*) or with* 5 *and* 6 *marked on the axis.*

A comparison and "year 6*" without that supporting work (for example a single row,* $B_6 > A_6$*) earns M1A1A0. The year may be implicit in the subscript (*$B_6 > A_6$ *with no year written still earns the year A1); it should be stated, so add a feedback note. Accept "after* 6 *years" for year* 6*, with a feedback note that the question asks for the year. A bare "year* 6*" with no comparison shown earns M0A0A0. "Year* 5.78*" is not a whole year: the intersection is supporting work, but the year A1 is lost (M1A0A1).*